

Course Outline: CSC101

Course Title: Introduction to Computer Programming

Course Code: CSC101

Description: Master the fundamentals of logic and problem-solving using Python. This course covers core syntax, data types, and control structures to build a solid foundation for modern software development.

Phase 1: Foundations & Data Handling

Focus: Getting comfortable with the environment and how computers store information.

- **Introduction to Python:** Setting up the IDE, understanding the interpreter, and writing your first "Hello World" program.
 - **Variables and Constants:** How to name and store data effectively.
 - **Data Types:** Deep dive into Integers, Floats, and Booleans.
 - **String Manipulation:** * Slicing and indexing strings.
 - Essential string methods (upper, lower, replace, find).
 - String formatting (f-strings).
 - **Basic Arithmetic Operators:** Using Python as a calculator +, -, *, /, //, %).
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Phase 2: Decision Making & Logic

Focus: Teaching the program how to make choices based on data.

- **Comparison Operators:** Understanding equality (==), inequality (!=), and relational operators (<, >, etc.).
 - **Logical Operators:** Combining conditions using and, or, and not.
 - **The if Statement:** Executing code based on a single condition.
 - **Branching with elif and else:** Handling multiple scenarios and providing default outcomes.
 - **Nested Conditions:** Placing conditions inside other conditions for complex logic.
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Phase 3: Iteration & Automation

Focus: Eliminating repetitive tasks through the power of loops.

- **Introduction to Loops:** Why we use iteration in programming.
- **The while Loop:** * Conditional iteration.
 - Avoiding infinite loops and managing loop counters.
- **The for Loop:** * Iterating over sequences (strings and ranges).
 - Using the range() function effectively.
- **Loop Control Statements:**
 - break: Exiting a loop prematurely.
 - continue: Skipping the current iteration.
- **Nested Loops:** Introduction to 2D patterns and basic matrix-style iteration.